

YASH HOODA

AI Engineer | Data Engineer | Software Engineer

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[LinkedIn](#) | [GitHub](#) | [Portfolio](#)

EDUCATION

University of Texas at Dallas

Bachelor of Science, Computer Science

Richardson, TX

May 2024

WORK EXPERIENCE

GSR Group — Data Engineer

Houston, TX

March 2026 – Present

- Built an end-to-end Toast POS API ingestion pipeline for 25+ Nothing Bundt Cakes franchise locations using Microsoft Fabric, Spark notebooks, and medallion architecture (Bronze/Silver/Gold).
- Engineered incremental ETL/ELT workflows to process restaurant sales, payments, refunds, voids, discounts, and operational transaction data at scale.
- Developed centralized data models and analytics pipelines powering downstream Power BI dashboards and Azure SQL reporting systems.
- Designed control tables and orchestration logic for automated backfills, scheduling, monitoring, and data quality tracking across multi-store franchise environments.
- Optimized large-scale ingestion workflows to handle Fabric compute capacity constraints, batching strategies, and scalable incremental processing.
- Worked with REST APIs, JSON payloads, Spark DataFrames, Delta Lake tables, SQL, Python, and cloud-based analytics infrastructure to support operational reporting and business decision-making.

Ternio Group — Healthcare Supply Chain Analytics — Data Engineer (Contractor)

Sugar Land, TX

September 2025 – February 2026

- Engineered automated Python + PySpark ETL pipelines for multi-hospital surgical instrument and sterilization datasets, producing clean, production-grade Lakehouse tables.
- Built scalable Microsoft Fabric dataflows integrating SQL, PowerQuery M, and Spark notebooks for anomaly detection, OR utilization, and compliance risk analysis.
- Developed real-time Power BI dashboards used by hospital leadership to monitor throughput, utilization, and multi-million-dollar compliance metrics.
- Improved data transformation efficiency and reduced manual reporting workload by 70%, replacing Excel and legacy processes with cloud-native automation.
- Optimized SQL queries and Lakehouse structures to improve performance across high-volume datasets (10M+ rows).

Archrock — Natural Gas Compression Services — Data Engineer Intern

Houston, TX

May 2025 – August 2025

- Automated phishing simulation and employee training metrics tracking using Power BI dashboards, reducing reporting delays by 50%.
- Used Microsoft Fabric, Lakehouses, and Azure Synapse pipelines to process unstructured cybersecurity training data into clean, analyzable tables.
- Developed logic to evaluate phishing click rates, training effectiveness, and data quality issues using Python and SQL.
- Collaborated with cybersecurity and IT leadership to present insights, contributing to safer internal communication protocols.

Century Instrument & Machine Company (CIMCO) — Data Analyst (Contractor)

Houston, TX

February 2025 – May 2025

- Conducted supplier performance analysis evaluating pricing, delivery times, and quality metrics, identifying 10% potential cost savings and optimizing procurement strategies.
- Analyzed and cleaned 1,000+ rows of operational data; developed automated reports and dashboards reducing reporting time by 30%.
- Designed a comprehensive machinist training document; researched and presented four cost-effective training software solutions, reducing training software expenses by 25%.

Outlier.AI — AI Trainer (Freelance)

Remote

May 2024 – February 2025

- Evaluated AI model outputs, chatbot responses, and Python/NumPy code for accuracy and quality, providing feedback that improved model performance by 90%+.
- Participated in NLP and AI prompt evaluation projects, developing deep expertise in LLM behavior, output quality assessment, and AI prompt engineering.

Parijat Controlware — Software Engineer Intern

Houston, TX

May 2023 – June 2023

- Developed automation logic using PLCs in a team environment.
- Applied object detection and machine learning using RoboFlow and Python for industrial automation use cases.

AI & PERSONAL PROJECTS

Personal Website RAG Chatbot with Memory

2025 – Present

Tools: Python, Claude API (Anthropic), LangChain, Vector Store (FAISS/Chroma), Streamlit, RAG, Embeddings, Persistent Memory

- Architected and deployed a full-stack RAG chatbot embedded in a personal portfolio website, enabling visitors to interactively query resume content, projects, and experience in natural language.
- Integrated the Claude API (claude-sonnet) as the core LLM backbone, leveraging its advanced reasoning and instruction-following for high-quality, context-aware responses.
- Implemented persistent vector store memory using FAISS/Chroma with LangChain embeddings, enabling multi-turn conversation recall and document retrieval across sessions.
- Designed a retrieval-augmented generation (RAG) pipeline that chunks, embeds, and indexes resume/portfolio content, then dynamically retrieves relevant context at inference time.
- Deployed publicly on Vercel with a Streamlit-powered chat UI, serving as a live demo of end-to-end AI application development.

HoodaAgents — AI Agent Framework

January 2025 – June 2025

Tools: Python, LangChain, OpenAI API, Streamlit, Tavily Search, Vector Store, RAG, NLP

- Designed and built a full-stack AI Agent framework capable of dynamic memory recall and personalized chatbot personas.
- Integrated LangChain and OpenAI API to handle task delegation, RAG pipelines, and persona-based conversation workflows.
- Deployed public demo with Streamlit UI and Tavily web search integration; included Hindi language support and vector store memory.
- Positioned as a customizable agent stack for future client deployments under the startup brand 'HoodaAgents'.

Liver Cancer Prediction — ML Model

February 2025

Tools: Python, Scikit-Learn, Pandas, NumPy, Matplotlib, Seaborn, EDA, Feature Engineering

- Analyzed a dataset of liver disease patient records; conducted EDA to identify key risk factors influencing liver cancer development.
- Implemented and optimized multiple ML algorithms (Random Forest, Logistic Regression), achieving strong predictive accuracy with interpretable feature importance visualizations.

Virtual TA Chatbot — UT Dallas

February 2024 – May 2024

Tools: Python, Rasa, NLP, LLM Integration, Visual Studio Code

- Collaborated within a 6-member team to design and develop a Virtual TA chatbot for the UT Dallas student portal.
- Engineered and trained conversational intents using Rasa; applied NLP and LLM integration to deliver real-time, context-aware academic support.

TECHNICAL SKILLS

- **AI / ML:** LangChain, OpenAI API, Claude API (Anthropic), RAG, Vector Stores (FAISS, Chroma), Embeddings, NLP, TensorFlow, Scikit-Learn, Prompt Engineering, LLM Fine-Tuning
- **Data Engineering:** PySpark, SQL, Microsoft Fabric, Databricks, ETL/ELT, Lakehouse Architecture (Bronze/Silver/Gold), Delta Lake, Medallion Architecture, REST API Ingestion
- **Programming:** Python (Pandas, NumPy, PySpark), SQL, JavaScript, Java, C/C++, R
- **Cloud & Platforms:** Azure (Synapse, Fabric, Foundry, Dataverse), Google Cloud, IBM Cloud, Microsoft Power Platform
- **BI & Visualization:** Power BI, DAX, PowerQuery M, Streamlit, Jupyter

- **Big Data:** Apache Spark, Spark DataFrames, Delta Lake, Data Pipelines, Batch & Incremental Processing
- **Tools:** Git, GitHub, VS Code, Postman, Databricks, Azure SQL
- **AI Tools:** Claude (Anthropic), ChatGPT (OpenAI), Gemini, Meta Llama, Ollama, DeepSeek R-1, Grok-3, GitHub Copilot, Microsoft Copilot

CERTIFICATIONS

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|---|---------------------|
| • Databricks Certified Data Engineer Associate — Databricks | Apr 2026 – Apr 2028 |
| • Microsoft Power Platform Fundamentals (PL-900) — Microsoft | April 2026 |
| • IBM Professional AI Engineering Certificate — Coursera | August 2024 |
| • TensorFlow Developer Certification — DeepLearning.AI | January 2024 |
| • IBM Professional Data Science Certification — Coursera | November 2024 |
| • Vanderbilt University AI Prompt Engineering Certification — Coursera | February 2025 |